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Introduction

SQL Views are a standard best practice for database administrators and developers. They can simplify complex queries, enforce security, and create reusable abstractions over underlying tables. This paper will review when one should use a SQL View, followed by a comparison of Views to Functions and Stored Procedures.

When to use a SQL View

SQL Views are most valuable when you need to present data in a consistent, simplified, or secure manner. Common uses include:

Convenience: As a saved Select statement or named query, Views enable developers to write the code once and run it repeatedly, despite often representing complex code. Views can pre-shape data for BI tools, reducing the need for complex transformations at query time.

Security: A View blocks off the ability to access tables directly and only gives access to the View. The View allows permissions to the public to be granted or denied. Views can expose only the necessary columns or rows (referred to as a vertical or horizontal partition), allowing users to query data without direct access to base tables and any confidential information within. Views coded with the security parameter “With Schemabinding” enforces database integrity by preventing the deletion of any table in which a View is dependent.

Base Views: It is considered standard practice to create a base View for every table within a database. This is otherwise known as an abstraction layer and done for security and ease of future updates. Views provide a stable interface over changing tables. If underlying tables evolve, a View can preserve backward compatibility for applications.

SQL Views, Functions, and Stored Procedures

Although Views, Functions, and Stored Procedures all encapsulate logic, they serve different purposes and have different strengths. Some comparisons and highlights of each are below.

Views are ideal for abstraction, security, and simplifying repeated queries. Views represent a virtual table based on a Select query. It cannot accept parameters and is best for read-only operations.

Scalar Functions are like a View in that it saved, repeatable code; however, Functions do accept parameters and can return a single value (scalar) or a result set (table-valued). Functions are useful for reusable calculations, formatting logic, or parameterized data retrieval. Functions are often used inside queries (a Select statement), Joins and/or Where clauses.

Stored Procedures, sometimes called sprocs or procs, are also a precompiled SQL code that can be saved. Stored Procedures accept parameters, can contain multiple SQL statements including branching logic, and can include transactions such as modifying data with Insert, Update or Delete statements. Stored Procedures are often used for business logic, ETL operations, or batch processing.

Summary

SQL Views are a foundational tool for simplifying data access, improving security, and creating stable interfaces for applications. While Views, Functions, and Stored Procedures all encapsulate saved, repeatable query logic, they serve distinct roles: Views for abstraction, Functions for parameterized logic, and Stored Procedures for procedural operations. Together, these SQL features help developers build maintainable, scalable, and efficient database systems.

Citation:

Root, R., 2021, Creating Views, Functions and Stored Procedures. Playlist retrieved May 20, 2026 at <https://www.youtube.com/playlist?list=PLfycUyp06LG8cefs0gA38wO7nFrRjD5Ad>.